Can a Single AI Data Center Be Seen in US Grid Demand?

A pre-registered detectability study of 60 sites and 14 balancing authorities

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> **Status: no finding yet.** This study measured what the public data *could* reveal. It has not measured any data center's actual effect on the grid. That test is the next stage, and it has not started.

In plain English

Imagine trying to tell, from your home's electricity bill alone, whether a neighbor plugged a space heater into your outlet. If your bill swings by hundreds of dollars with the seasons, a heater adds too little to notice. If your bill barely moves, you would spot it right away.

The US grid works the same way. Each regional grid operator, called a **balancing authority**, publishes its electricity demand every hour. AI data centers now draw hundreds of megawatts, and some more than a gigawatt. People often assume that this new demand must be obvious in the data. But a heat wave can add far more demand than a data center does.

So before asking "did this data center change demand?", this study asks a prior question: **if it did, could we even tell?** It found three things:

1. **Mostly no, for one site at a time.** Of 60 listed US sites with power by October 2026, only **2** pass the test. Together they hold 345 of the 15,835 MW in the list.
2. **Size of the grid matters more than size of the campus.** A 147 MW site in New Mexico can be seen. The 1,244 MW New Carlisle campus cannot be seen in PJM's total, because PJM's demand is so large that its normal swings would hide it.
3. **The rules were written first.** The method was committed to git before any model ran, so it could not be adjusted to reach a preferred answer.

Abstract

Attributing electricity demand growth to AI data centers requires knowing how small a change hourly grid data can reveal. We estimate a minimum detectable step for 11 demand series from EIA-930 for 2019-2026, using an untuned, weather-normalized LightGBM baseline scored strictly out of sample and only on windows that end before each balancing authority's first listed site. Of 21 balancing authorities holding sites listed by Epoch AI, 14 met a pre-registered 2% share rule, and 10 of those had enough history for an estimate. Detection limits ran from 1.0% to 5.5% of 12-month peak demand. Two sites without disqualifying flags cleared their limits: Meta Kuna (198 MW against 98 MW in Idaho Power) and Meta Los Lunas (147 MW against 77 MW in PNM). No eligible site cleared

its limit in PJM, ERCOT, or Southern Company, where the limits were 3,803, 2,675, and 1,678 MW. In a secondary analysis, New Carlisle (1,244 MW) cleared the 590 MW limit of PJM's smaller AEP zone. These results support a narrow event study in small balancing authorities and suggest that single-campus claims at the level of a large market need stronger evidence. No effect of any site has been estimated.

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1. Introduction

Demand from AI data centers is a common explanation for rising load forecasts. Testing that claim in observed data is harder than it looks. Hourly demand moves with weather, the calendar, and other customers, and any new campus arrives on top of those movements. An event study applied to a site that is too small to detect gives a null result that means nothing, and a positive result could easily be noise.

Statisticians handle this by calculating power before running a test. This paper does that for grid demand. It asks what size of sudden, lasting step in demand each series could reveal, using a 5% false positive rate and 80% power, and it compares that limit with each listed site.

Three design choices protect the result:

- **Pre-registration.** The analysis plan was committed before any model was fit. Later changes are dated amendments.
- **No look at the real effect.** Every number comes from windows that end before a balancing authority's first listed site. No residual has been computed on or after a real start date.
- **Untuned model.** The LightGBM settings are common starting values, fixed before the first fit. Tuning them against these windows would shrink the limits and flatter the result.

2. Data

Input	Source	Coverage	Use
Hourly demand and day-ahead forecast	EIA-930	2,861,089 rows, 21 BAs, 2019-01-01 to freeze	Model target and benchmark
PJM's AEP zone demand	EIA-930 sub-BA	68,354 rows	Secondary analysis
Site power, dates, construction notes	Epoch AI	77 US sites, 60 with power by freeze	Selection and comparison
Hourly temperature and humidity	Open-Meteo	8,308,344 rows, 121 counties	Weather features
Population and centroids	US Census Bureau	All counties and ZIP areas	Weather weights, site location
BA boundaries	Esri layer built on EIA data	60 polygons	Lookups only. BA codes are stored, not polygons

Site locations. Of the 60 sites, 29 were placed by the Census geocoder, 19 by a ZIP centroid, 7 by Google Earth links in Epoch's notes, and 5 by a hand-entered ZIP with a citation. None were left unlocated. Where both the geocoder and Earth links were available, on 4 sites, they agreed on the BA.

Disqualifying flags. Seven sites carry an on-site power flag, because power generated at the site may never appear in grid demand. Six come from a text search of Epoch's notes. One, Colossus 1, was added by hand with a citation. Ten sites sit inside more than one BA polygon and cannot satisfy the go test, because the polygons do not say which BA serves them.

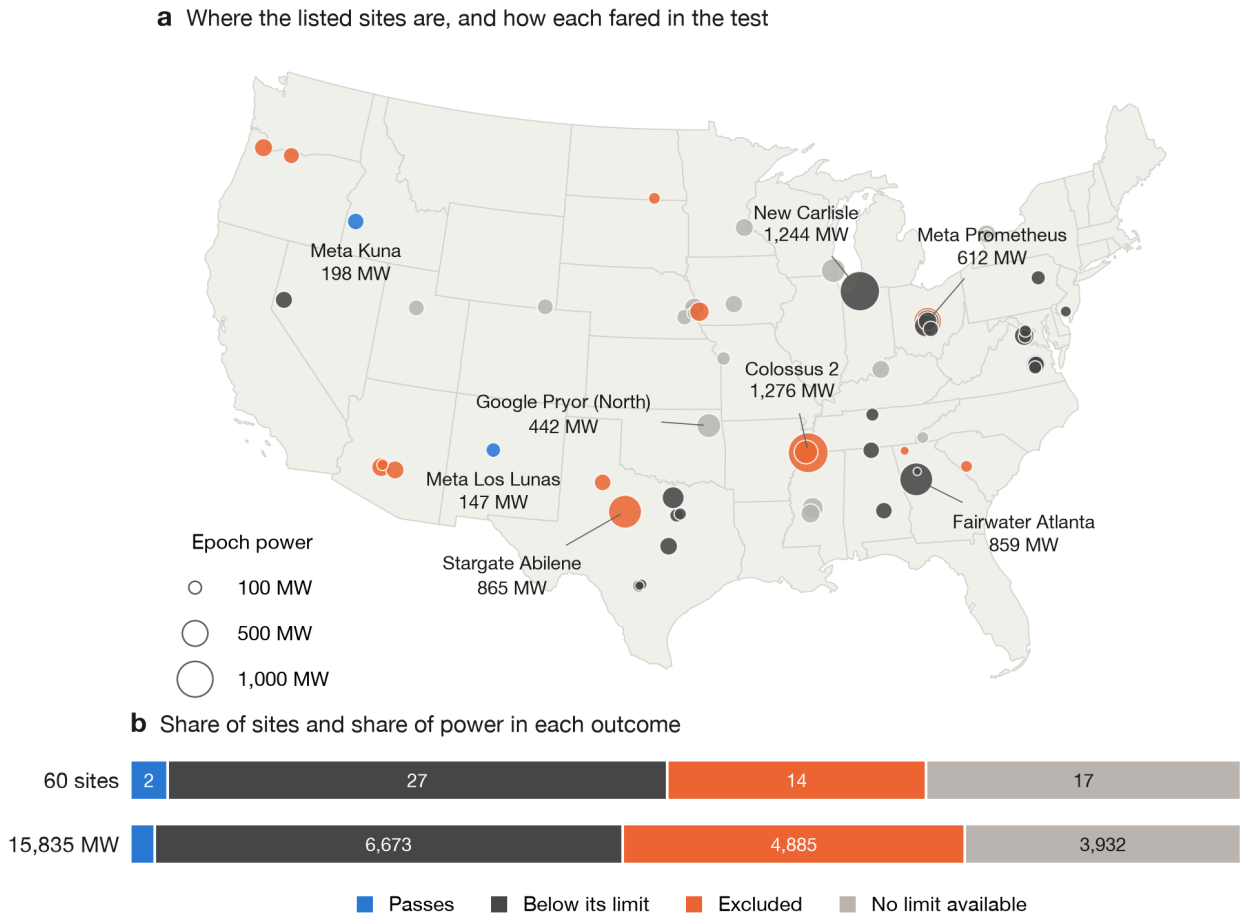


Figure 1. Where the listed sites are, and how each fared. a, Each circle is one site, sized by Epoch's estimated power. Blue sites pass. Dark gray sites are smaller than their BA's limit. Orange sites are excluded for an on-site power flag or overlapping BA areas. Light gray sites have no limit, because their BA missed the 2% rule or has too little history. **b,** The two passing sites are 2 of 60 sites and 345 of 15,835 MW.

Why this is hard: demand moves far more than a campus adds

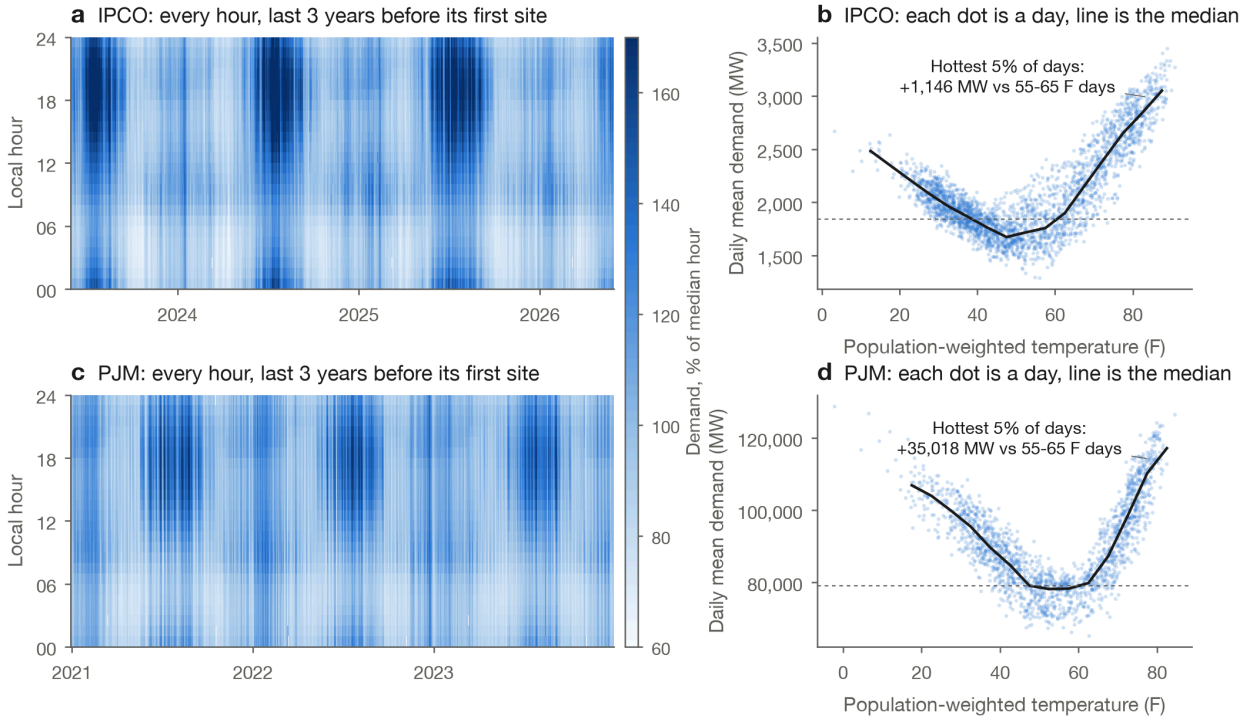


Figure 2. The normal rhythm that a data center has to stand out from. a, c, Each vertical stripe is one day and each row is one local hour, so a dark region means high demand. Summer afternoons and evenings stand out in both grids. **b, d,** Demand against temperature forms a U, rising on cold days and hot days. The hottest 5% of days add 1,146 MW in IPCO, about 6 times Meta Kuna's 198 MW. They add 35,018 MW in PJM, about 28 times New Carlisle's 1,244 MW. All data predates each BA's first listed site.

3. Methods

Candidate selection. A BA is a candidate if its listed site power is at least 2% of its 12-month peak hour. Site power is Epoch's `Power (MW)` on the latest timeline row at or before the freeze. Rows after the freeze are projections and set neither power nor dates.

Baseline. Trees cannot extrapolate a trend, and a flat trend would look like new load. So a linear trend is fit on the training window, LightGBM predicts the detrended demand, and the trend is added back. Features are local hour, day of week, month, US federal holiday, temperature, humidity, and cooling and heating degrees from a 65 F base. Weather is weighted by population across up to 10 counties. There is no lagged demand, because lags would absorb the very step being measured. Hours at or below zero, or above 3 times the BA's median, are dropped and counted.

Step estimate. Test dates t fall every 30 days. For each one, the model trains on data up to $t - 365$ days and predicts the next 545 days. With daily mean residual r_d:

$$e(t) = \frac{1}{180}\sum_{d=t}^{t+179} r_d \; - \; \frac{1}{365}\sum_{d=t-365}^{t-1} r_d$$

The last 180-day window must end before the BA's first listed site, so the real effect stays out.

Detection limit. With no real step in these windows, the values of $e(t)$ describe the noise. A step is called detected when $e(t)$ exceeds the 95th percentile of that noise. Adding S MW after t raises $e(t)$ by exactly S without changing the model, so:

$$\text{minimum detectable step} = q_{0.95}(e) - q_{0.20}(e)$$

This is the smallest step found in 80% of windows. BAs with fewer than 12 test dates get no number.

Go test. A site passes if it has no on-site power flag, sits in one BA polygon, belongs to a candidate BA with a limit, and has power at or above that limit.

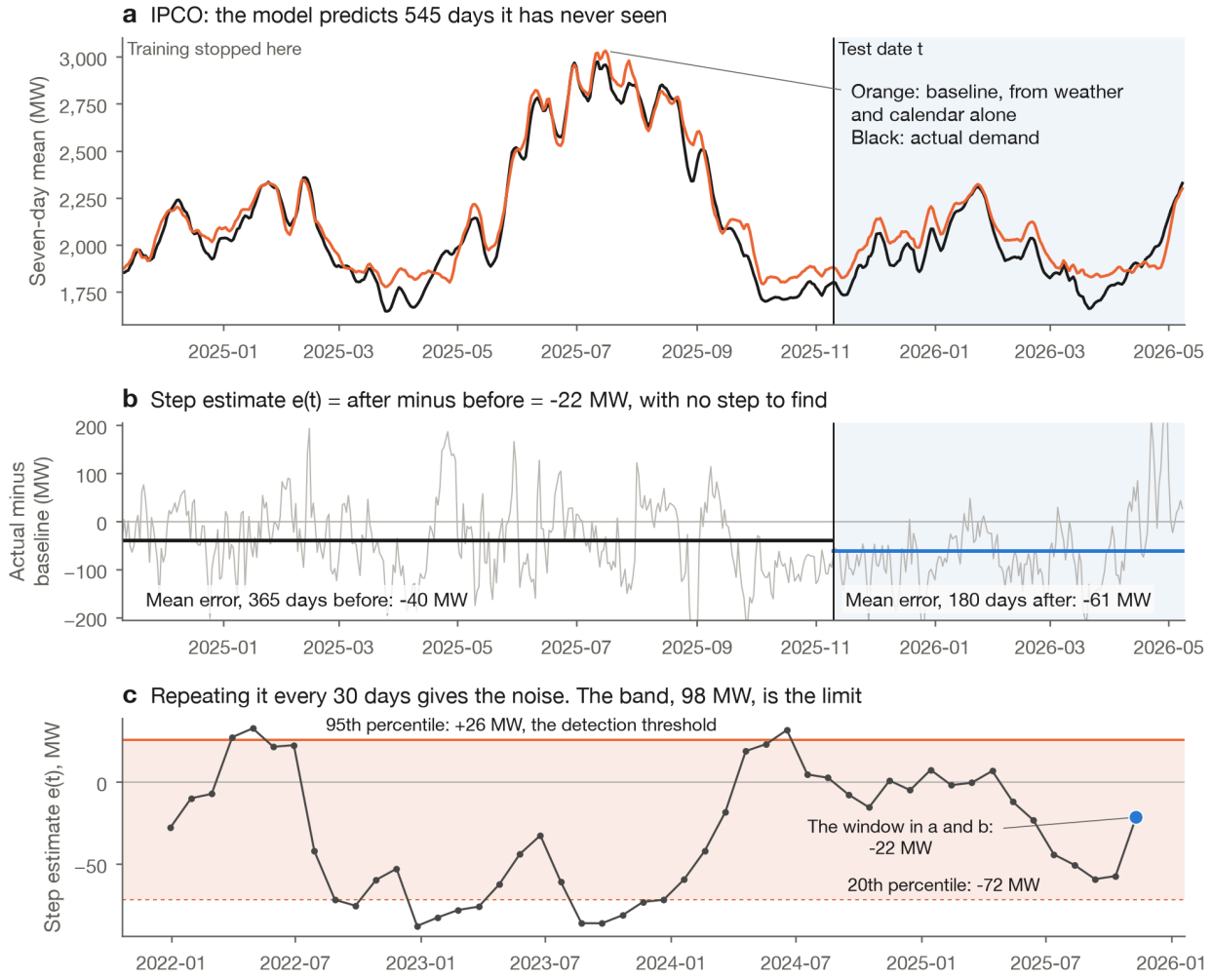


Figure 3. The method on one real window. a, For IPCO's newest test window, the baseline tracks demand across 545 days it never saw. **b,** The daily error is averaged over the year before the test date and the 180 days after it. The difference is one step estimate: –22 MW, even though no step was added. **c,** Repeating this every 30 days shows how far the estimate wanders by chance. The orange band, from the 20th to the 95th percentile, is 98 MW wide. That width is IPCO's detection limit.

4. Results

4.1 Which grids could be tested

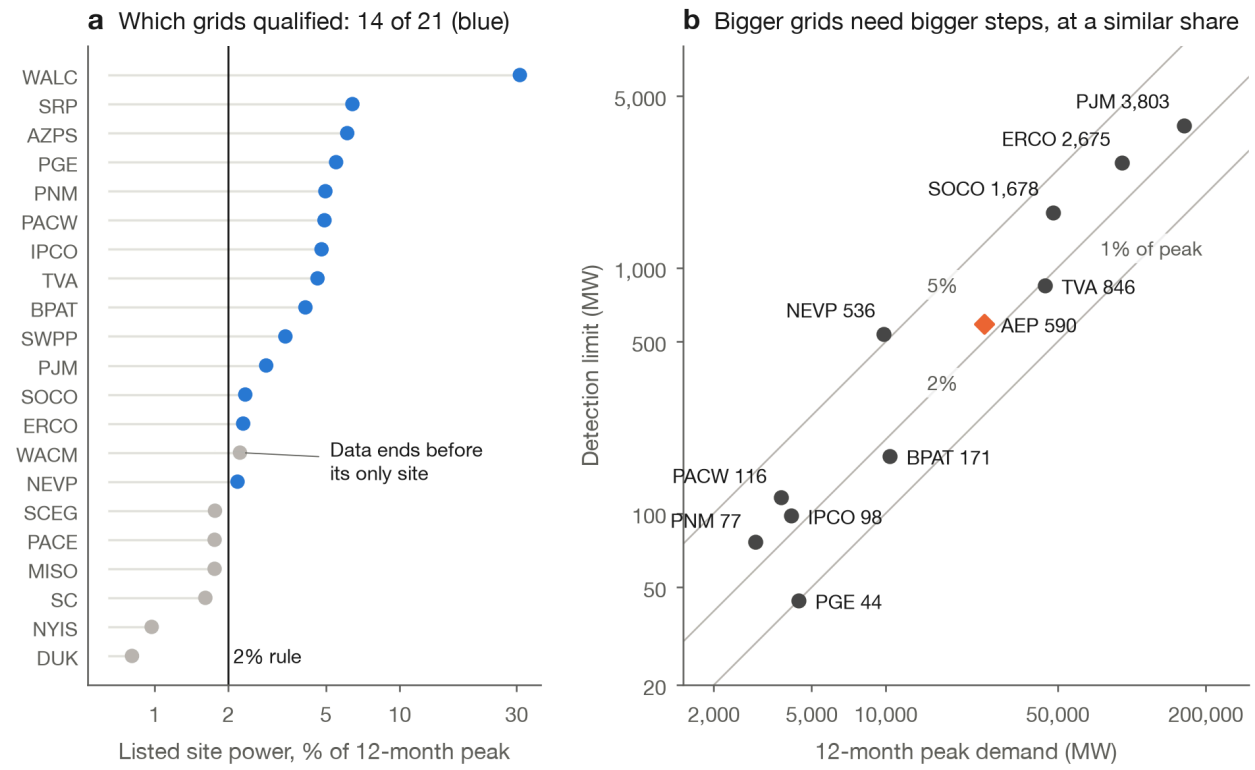


Figure 4. Selection, and how limits scale with grid size. a, 14 of 21 BAs meet the 2% rule. WACM passes on share but is out, because its data ends on 2026-04-02 and its only site starts later. MISO misses at 1.76%. WALC's 31% comes from three Phoenix-area sites inside overlapping polygons. **b,** Limits rise with grid size at about the same share of peak, between about 1% and 5.5%. Both axes are logarithmic. The orange diamond is PJM's AEP zone.

SWPP (7 test dates), AZPS, SRP, and WALC (10 each) had too little history before their first site to get a number, which leaves 10 BAs and the AEP zone.

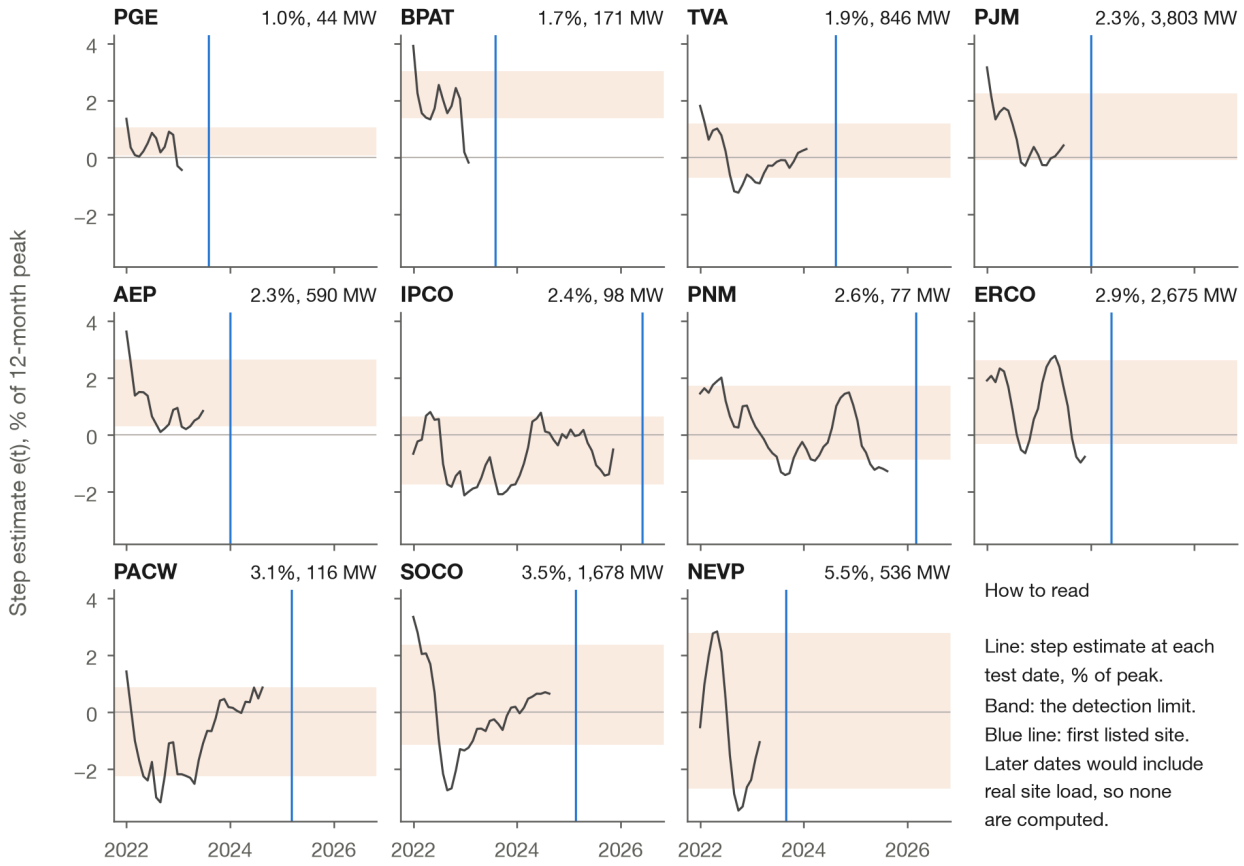


Figure 5. The noise in every series, on one scale. Series run from the smallest limit to the largest, as a share of peak. Headers give the limit as a share of peak and in MW. Lines stop 180 days or more before the blue line by design. Some series start high: BPAT, PJM, AEP, and SOCO estimates exceed 3% of peak in early 2022, and AEP's values are all positive, so its demand was already beating a linear trend before its first listed site. Later growth can look like a step, which is one reason a step alone cannot prove a cause.

4.2 Site by site

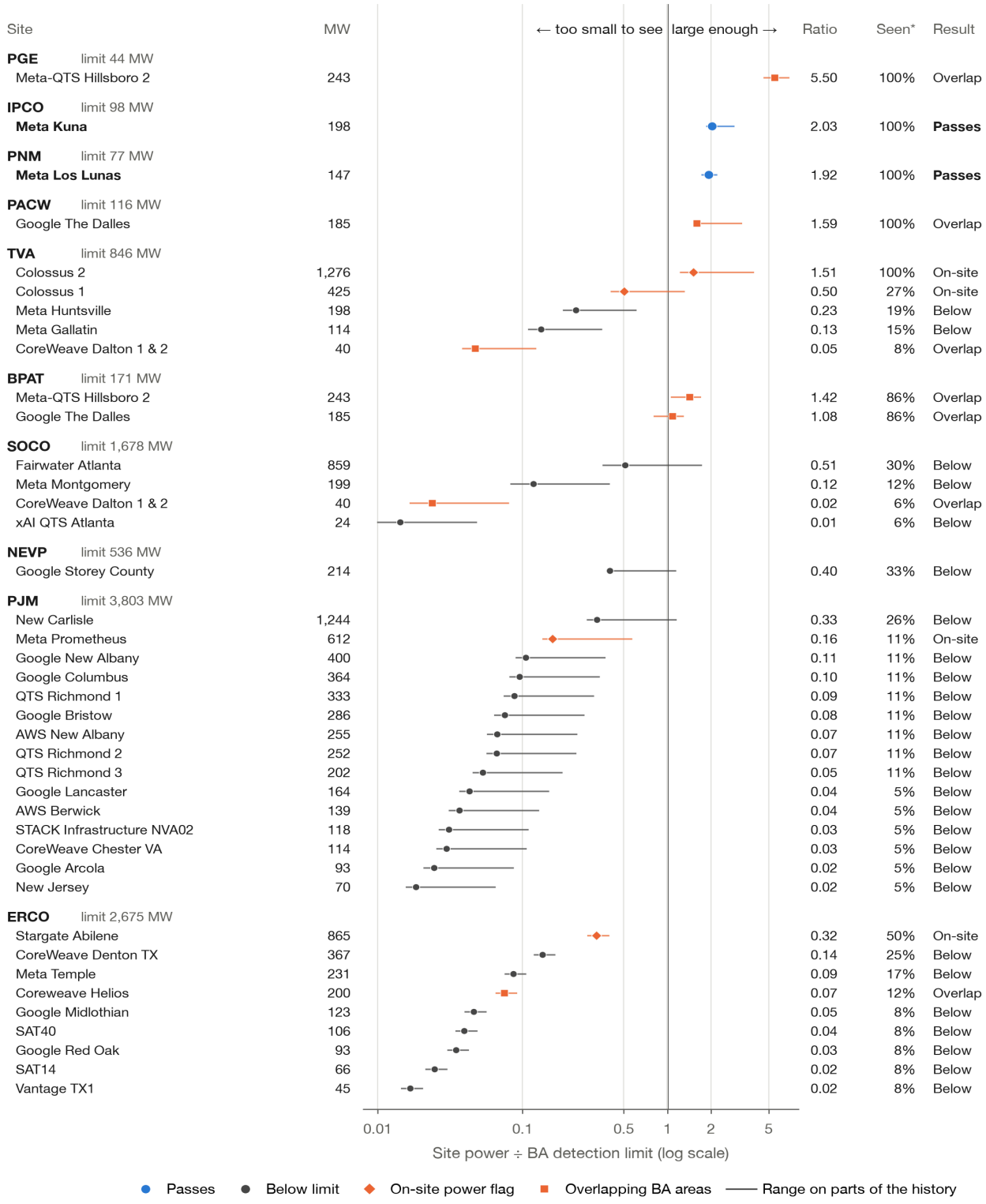


Figure 6. Every site against its BA's limit. Read this like a table with a picture in the middle. A point right of the vertical line is a site larger than its BA's limit. Each line shows how far that ratio moves when the limit is recomputed on parts of the history (Section 5). A site in two BA areas appears twice. *Seen* is the share of test windows in which a step of the site's full power would cross the threshold. It is descriptive and not part of the go test.

Table 1 lists the sites that decide the result.

Site	BA	Power (MW)	Limit (MW)	Ratio	Result
Meta Kuna	IPCO	198	98	2.03	**Passes**
Meta Los Lunas	PNM	147	77	1.92	**Passes**
Meta-QTS Hillsboro 2	PGE	243	44	5.50	Large enough, excluded for overlap
Colossus 2	TVA	1,276	846	1.51	Large enough, excluded for on-site power
Fairwater Atlanta	SOCO	859	1,678	0.51	Below limit
New Carlisle	PJM	1,244	3,803	0.33	Below limit
Stargate Abilene	ERCO	865	2,675	0.32	Below limit, and on-site power

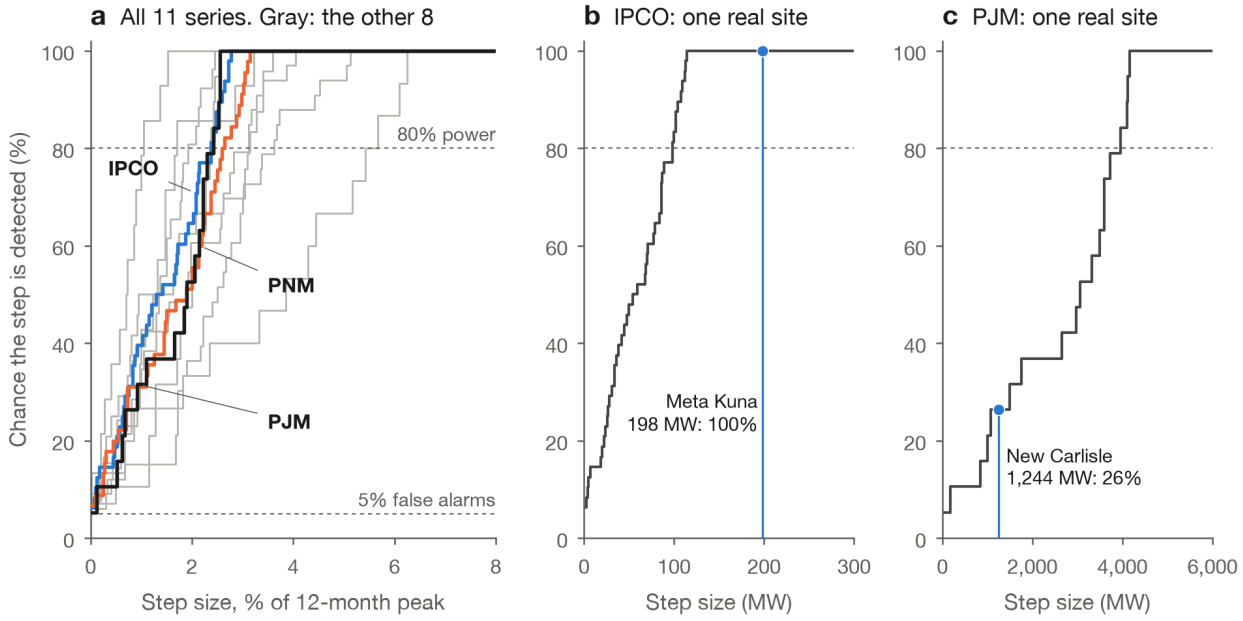


Figure 7. How the chance of detection grows with step size. a, Each curve starts near the 5% false alarm rate and rises as the step grows. Most reach 80% between 1% and 4% of peak. **b,** Meta Kuna's 198 MW would be detected in all of IPCO's windows. **c,** New Carlisle's 1,244 MW would be detected in 26% of PJM's windows.

4.3 Secondary analysis: PJM's AEP zone

AEP is much smaller than PJM, with a 12-month peak of 25,122 MW and a limit of 590 MW. New Carlisle's 1,244 MW clears it, needing 47% of its power to be seen. By the plan's rule, this analysis cannot produce a go by itself. It uses PJM's weather counties because the zone has no polygon of its own, so its limit may be too high.

4.4 How good is the baseline?

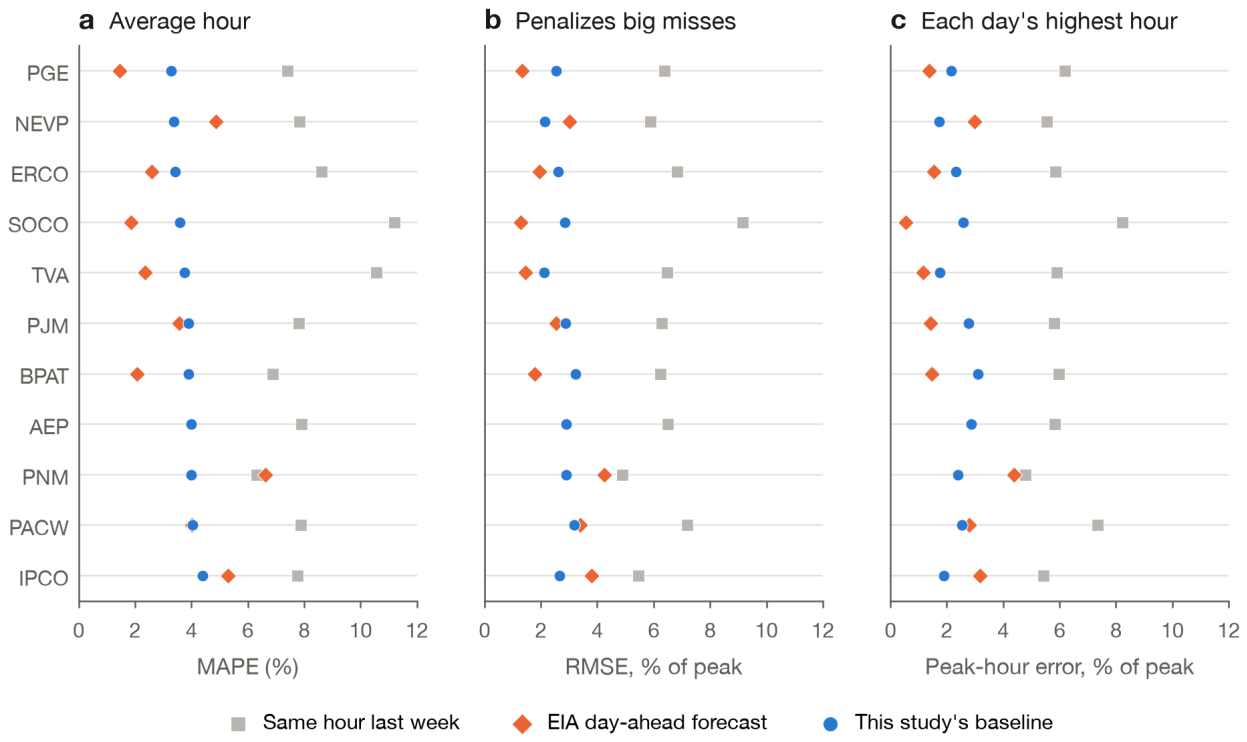


Figure 8. The baseline against two benchmarks. Lower is better. The baseline beats a same-hour-last-week guess in every series and on every measure. On average error, it also beats EIA's day-ahead forecast in PNM, IPCO, and NEVP and ties in PACW, and it loses elsewhere. This is not a fair forecasting contest: the baseline is given actual weather, while EIA forecasts a day ahead with fresher demand data. AEP has no EIA forecast.

5. Robustness

Check	Result
Rerun at a new commit	Identical limits for all 15 series
Full rebuild from a fresh clone and empty lake	Results file matched on every line but the commit ID
Step injected into real demand	Detected in 77-80% of windows at the limit
No step added	5-8% of windows crossed the threshold
Limit recomputed on parts of each history (not pre-registered)	Both passing sites still clear it

The sensitivity check was added after the first complete run and plays no part in the go test. IPCO's limit ranged from 69 to 106 MW and PNM's from 67 to 85 MW, against sites of 198 and 147 MW. Los Lunas was at 114 MW until 2026-10-26, which also clears 85 MW. Other limits are far less stable. SOCO's ranged from 503 to 2,388 MW, and PJM's from 1,086 to 4,434 MW. New Carlisle sits inside PJM's range, so "below limit" there is a best estimate and not a firm boundary. The full table covers every series.

6. Discussion

The main result is about grids, not campuses. Limits scale with the grid, so the same campus can be easy to see in one place and invisible in another. A claim that one campus is visible in a total for PJM or ERCOT should be read with care, since these limits exceed every listed site in those markets.

The result also shows where a test could work. The strongest case in the data is Meta-QTS Hillsboro 2, at 5.5 times the PGE limit with data since 2023. It is excluded only because the polygons overlap, so citing its serving utility could add it to the next stage through a dated amendment.

7. Limitations

- **Could, not did.** This study says what could be detected, not what happened.
- **Few independent windows.** Each estimate spans 545 days and test dates are 30 days apart, so a BA has far fewer independent windows than dates. Every limit is rough.
- **Sudden steps only.** Real campuses ramp up over months, which is harder to detect.
- **Estimated site data.** Epoch's power and dates are estimates from satellite imagery and permits, not metered load. A campus may draw well below its figure. The list is incomplete, and a first date can lag a real start. Epoch says Google Storey County's Building 1 was first operational in 2021, but its first row with power is 2023-08-30.
- **Existing load.** Both passing sites are on Meta campuses in small BAs, and a result there says little about PJM or ERCOT. Other large loads in the history widen the noise.
- **On-site power.** Text search only finds what Epoch wrote, so some flags may be missing. Meta Kuna's notes mention an offsetting solar array. Off-site solar would not reduce BA demand, but its connection should be confirmed before the next stage.
- **Geography.** BA polygons overlap and do not match county lines. 24 of 60 sites rest on ZIP centroids, and weather is a county-centroid approximation. SRP's weather is one county.
- **Timing.** Meta Kuna's first date is 2026-06-01, so a 180-day window is not complete until about 2026-11-28, after this freeze.
- **Data quality.** EIA-930 revises recent hours and has gaps and spikes. WACM stops on 2026-04-02.
- **Licensing.** The BA polygons are under the Esri Master License Agreement, not an open license. Anyone reproducing this depends on Esri keeping that layer available.

Process note. During this work, a run began against uncommitted code when a commit failed. It was stopped in under a minute and its output was deleted unread, before the code was committed and run. The build log records this.

8. Conclusion and next steps

Single AI campuses can be detected in small balancing authorities and generally cannot in large markets. The gate decision, recorded in ADR-0002, is a **go with a narrower next stage**:

1. Run the event study on IPCO and PNM, with placebo dates, placebo BAs, and 90% intervals, using a fresh snapshot once Kuna has 180 days.

2. Before computing anything at a real start date, commit an amendment fixing those details and citing the serving utility for overlapping sites, starting with Hillsboro.
3. Keep New Carlisle in the AEP zone as a labeled secondary result.
4. Have the method and results reviewed independently.

A null result will be published as a null result.

Reproduce this

You need git, make, uv, and Docker with Compose v2. On macOS with Colima, run `colima start` after a reboot and give the VM at least 4 GB.

```
git clone https://github.com/VijayChamakuri/ai-power-atlas.git
cd ai-power-atlas
make setup
```

`make setup` creates `.env`. Add a free EIA API key there. `make sources-check` also needs a free Census key and a contact address for SEC EDGAR.

```
make up
make ci
make pipeline
```

Command	What it does
`make up`	Starts Postgres, MinIO, Airflow, and MLflow
`make ci`	Runs ruff, `mypy --strict`, and the tests, with network access blocked
`make pipeline`	Pulls every source, runs the analysis, and draws the figures. About an hour
`make sources-check`	Rewrites `docs/source_status.md`. Exits 1 while LBNL and FCC block scripted requests

Local services are MLflow at http://localhost:5001, Airflow at http://localhost:8080, and the MinIO console at http://localhost:9001. All ports bind to loopback. Default local passwords are in `.env.example`.

`make pipeline` pulls whatever the sources serve that day. Providers revise history and Epoch updates its files, so later runs can differ. To work from the exact inputs behind this paper, download `atlas-m1-frozen-lake.tar.gz` from the data release into the repo folder, then run:

```
make lake-restore
make analysis
```

The restore checks a hash for every file and writes nothing if any file is damaged. `make paper` rebuilds the PDF, and needs Google Chrome. docs/log.md records the rebuild checks and input hashes.

Repository map

Path	Contents
docs/paper.pdf	This paper as a PDF
docs/analysis_plan.md	Pre-registered method and dated amendments
docs/m1_detectability.md	Generated results for every BA and site
docs/adr/	The go or pivot decision, and why history is never squashed
docs/DATA_SOURCES.md	URLs, licenses, attribution, and known issues
docs/log.md	Build log, data hashes, and differences from the plan
docs/PLAN.md	Full project plan
`src/atlas/ingest/`, `contracts/`, `lake/`	Source pulls, schema checks, and storage
`src/atlas/ml/`	Features, baseline, candidates, detectability, report, and figures
`dbt/seeds/`	Hand-entered rows, each with a citation
`tests/`	Tests using recorded real responses and slices of real data

Glossary

Term	Meaning
Balancing authority (BA)	An organization that keeps supply and demand matched in one part of the grid
Megawatt (MW)	A unit of power. This study's sites run from 24 to 1,276 MW
Baseline	The demand expected from weather and the calendar alone
Residual	Actual demand minus the baseline
Step	A sudden, lasting rise in demand, such as a new campus switching on
Detection limit	The smallest step found in 80% of windows, also called the minimum detectable effect
False positive	Calling a step detected when there is none. This study allows 5%
Power	The chance of detecting a step that is really there. This study targets 80%
Pre-registration	Committing the method before seeing results
Out of sample	Scored on data the model never saw while learning
Forest plot	A chart that lines up many estimates and their ranges against one reference line

References

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8. Rougier, Droettboom, and Bourne (2014). Ten Simple Rules for Better Figures. *PLOS Computational Biology*.

Code is under the MIT License. Data stays under each source's own terms.